Fluid Kinematics

Unit III - Mechanical Engineering

ME10008

Comprehensive Study Notes

Contents

1 Introduction to Fluids

1.1 What is a Fluid?

Definition

A substance which can flow freely when not in equilibrium condition is called a fluid.

States of Matter: Solid and Fluid (Liquid and Gas)
Characteristics of Fluids:

1. Has no specific shape of its own - conforms to the shape of the containing vessel
2. A small shear force will cause it to deform continuously as long as the force exists

2 Properties of Liquids

These properties are crucial as hydraulics mainly deals with water and other liquids.

2.1 Density (ρ)

Definition

Mass or weight per unit volume.

Unit: kg/m^3

$$\rho = \frac{\text{Mass}}{\text{Volume}}$$

2.2 Specific Gravity (S)

Definition

Ratio of specific weight of a liquid to the specific weight of a standard liquid (usually water).

$$S = \frac{\text{Specific weight of liquid}}{\text{Specific weight of water}}$$

Dimensionless quantity.

2.3 Viscosity (μ)

Definition

Resistance to shear stress between consecutive layers of fluid.

Unit: N·s/m^2 or Pa·s
Physical meaning: Internal friction in fluids that opposes relative motion between layers.

2.4 Cohesion

Definition

Tendency of liquid particles to remain as one assemblage due to intermolecular attraction.

2.5 Adhesion

Definition

Attraction between molecules of liquid and solid boundary surfaces.

2.6 Surface Tension

Definition

Property caused by force of cohesion at the free surface of a liquid.

Effect: Creates a "skin" on the liquid surface that resists external force.

2.7 Capillarity

Definition

Rise or fall of liquid in a narrow tube due to combined effect of cohesion and adhesion.

Example: Water rises in a glass tube (adhesion ¿ cohesion), while mercury falls (cohesion ¿ adhesion).

2.8 Compressibility

Definition

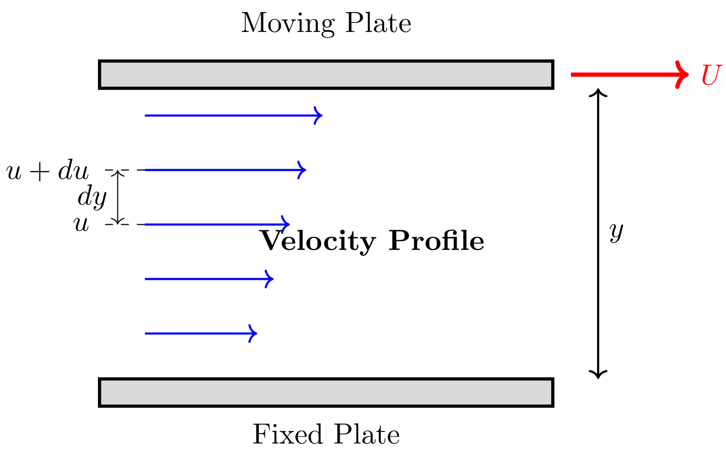
Change in volume of a fluid due to external force (pressure).

Note: Liquids are nearly incompressible; gases are highly compressible.

3 Viscosity and Newton's Law

3.1 Newton's Law of Viscosity

When two layers of fluid at distance 'dy' apart move with different velocities (u and u+du), viscosity causes shear stress between layers.



Newton's Law of Viscosity:

$$\tau = \mu \frac{du}{dy}$$

where:

- τ = Shear stress ($\mathrm{N/m^2}$)
- μ = Dynamic viscosity ($\mathrm{N{\cdot}s/m^2}$)
- $\frac{du}{dy}$ = Velocity gradient or rate of shear deformation ($\mathrm{s^{-1}}$)

3.2 Types of Fluids Based on Viscosity

3.2.1 1. Newtonian Fluids

Definition

Fluids that follow Newton's law of viscosity. Viscosity (μ) does not change with rate of deformation.

Examples: Water, kerosene, air, most gases and simple liquids
Behavior: Linear relationship between shear stress and velocity gradient.

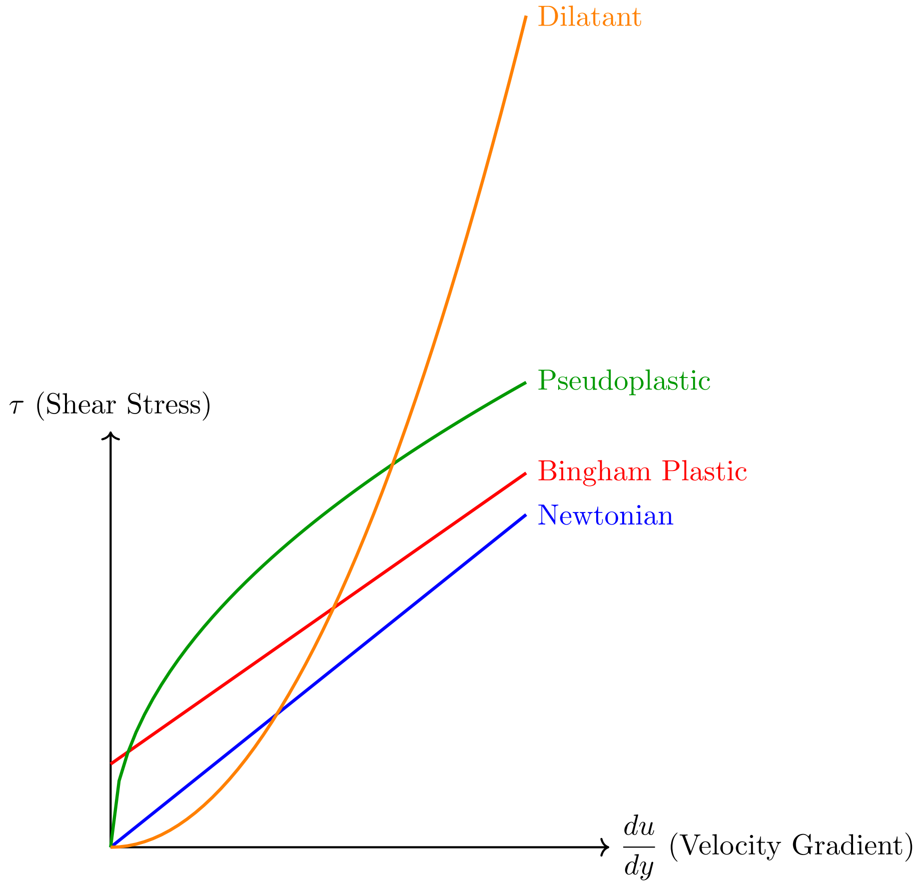
3.2.2 2. Non-Newtonian Fluids

Definition

Fluids that do not follow linear relationship between shear stress and rate of deformation.

Examples: Solutions/suspensions (slurries), mud flows, polymer solutions, blood, toothpaste
Study: These fluids are studied under *rheology* - the science of deformation and flow.

3.3 Classification of Non-Newtonian Fluids



Pseudoplastic: Shear thinning (paint)
Dilatant: Shear thickening (cornstarch solution)

4 Fluid Kinematics - Fundamentals

4.1 What is Kinematics?

Definition

Kinematics is the study of motion. Fluid kinematics describes fluid motion without considering the forces responsible for flow.

Focus of Study: Displacement, velocity, and acceleration of fluid particles.

Note: Fluid flow is always caused by external forces (pressure difference, gravity, etc.), but kinematics focuses only on the motion itself.

4.2 Two Methods of Studying Fluid Motion

Lagrangian Method
(Particle tracking)

↓

Eulerian Method
(Field variables at fixed points)

5 Lagrangian Method

Concept

Track the motion of individual fluid particles by specifying their coordinates as a function of time.

5.1 Path Line

The line connecting all points a particle passes through over time is called a **path line** or **Lagrangian particle path**.

Initial position at $t = t_0$:

$$\vec{r}_0 = \hat{i}x_0 + \hat{j}y_0 + \hat{k}z_0$$

5.2 Complete Description

Position vector as function of time:

$$\vec{r} = F(\vec{r}_0, t)$$

In components:

$$x = f_1(x_0, y_0, z_0, t)$$
$$y = f_2(x_0, y_0, z_0, t)$$
$$z = f_3(x_0, y_0, z_0, t)$$

Velocity: Obtained by differentiating position with respect to time

$$\vec{V} = \frac{d\vec{r}}{dt}$$

5.3 Disadvantages of Lagrangian Method

- Large number of fluid particles in actual systems
- Tracking each particle is time-consuming
- May take days or months for small quantities of fluid
- Becomes laborious and impractical for most engineering problems

6 Eulerian Method

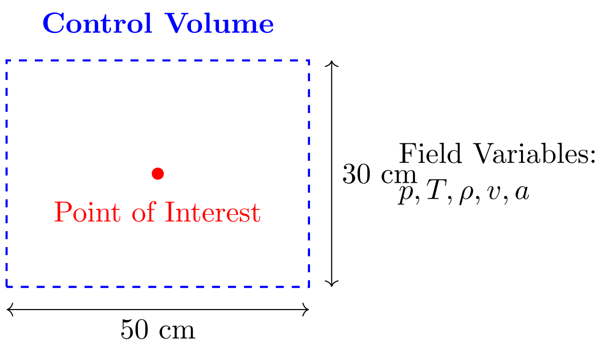
Concept

Select a point of interest (control volume) and define field variables at that location. Study how these variables change with time.

6.1 Field Variables

Field variables can be:

- Pressure (p)
- Temperature (T)
- Density (ρ)
- Velocity (v)
- Acceleration (a)



6.2 Mathematical Description

General form:

$$\text{Field Variable} = f(\text{space}, \text{time})$$

For velocity field:

$$\vec{V} = f(x, y, z, t)$$

In components:

$$u = g_1(x, y, z, t)$$
$$v = g_2(x, y, z, t)$$
$$w = g_3(x, y, z, t)$$

where u, v, w are velocity components in x, y, z directions.

6.3 Solved Example

Example

Given velocity field:

$$\vec{V} = (3x + 2y)\hat{\imath} + (2z + 3x^2)\hat{\jmath} + (2t - 3z)\hat{k}$$

Determine:

1. Velocity components u, v, w
2. Speed at point (1, 1, 1)
3. Speed at t=2s at point (0, 0, 2)
4. Classify the flow

Solution:
(i) Velocity components:

$$u = 3x + 2y$$
$$v = 2z + 3x^2$$
$$w = 2t - 3z$$

(ii) Speed at point (1, 1, 1):
Substituting $x = 1, y = 1, z = 1$:

$$u = 3(1) + 2(1) = 5$$
$$v = 2(1) + 3(1)^2 = 5$$
$$w = 2t - 3(1) = 2t - 3$$

$$V^2 = u^2 + v^2 + w^2 = 25 + 25 + (2t - 3)^2 = 4t^2 - 12t + 59$$
$$V = \sqrt{4t^2 - 12t + 59}$$

(iii) Speed at t=2s at point (0, 0, 2):
Substituting $t = 2, x = 0, y = 0, z = 2$:

$$u = 3(0) + 2(0) = 0$$
$$v = 2(2) + 3(0)^2 = 4$$
$$w = 2(2) - 3(2) = -2$$

$$V = \sqrt{0^2 + 4^2 + (-2)^2} = \sqrt{20} = 4.472 \text{ m/s}$$

(iv) Classification:

- **Unsteady flow** (velocity depends on time t)
- **Non-uniform flow** (velocity varies with position)
- **Three-dimensional flow** (velocity has components in all three directions)

7 Types of Fluid Flow

7.1 1. Steady and Unsteady Flow

7.1.1 Steady Flow

Definition

Flow in which pressure, temperature, velocity, etc. do not change with time at a given point.

$$\frac{\partial p}{\partial t} = 0, \quad \frac{\partial T}{\partial t} = 0, \quad \frac{\partial v}{\partial t} = 0$$

Steady Flow

$t = 10$s	$t = 20$s
$p = 10$ bar	$p = 10$ bar
$T = 400$K	$T = 400$K
$v = 45$ m/s	$v = 45$ m/s

Properties unchanged with time

7.1.2 Unsteady Flow

Definition

Flow in which pressure, temperature, velocity, etc. change with time at a given point.

$$\frac{\partial p}{\partial t} \neq 0, \quad \frac{\partial T}{\partial t} \neq 0, \quad \frac{\partial v}{\partial t} \neq 0$$

Unsteady Flow

$t = 10$s	$t = 20$s
$p = 10$ bar	$p = 15$ bar
$T = 400$K	$T = 450$K
$v = 45$ m/s	$v = 50$ m/s

Properties change with time

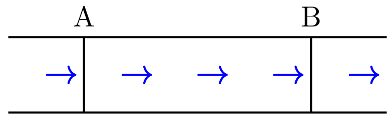
7.2 2. Uniform and Non-Uniform Flow

7.2.1 Uniform Flow

Definition

Flow in which velocity and other properties do not change from point to point at any given instant.

$$\frac{\partial v}{\partial s} = 0$$



Section A
$v = 45$ m/s

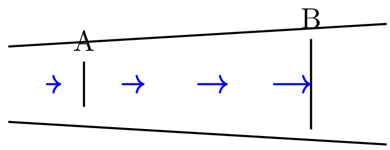
Section B
$v = 45$ m/s

7.2.2 Non-Uniform Flow

Definition

Flow in which velocity changes from point to point at any given instant.

$$\frac{\partial v}{\partial s} \neq 0$$



Section A
$v = 45$ m/s

Section B
$v = 70$ m/s

7.3 3. Laminar and Turbulent Flow

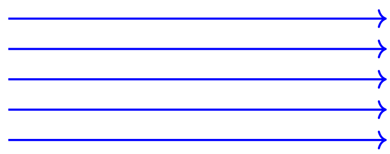
7.3.1 Laminar Flow

Definition

Fluid particles flow in defined, straight, parallel paths (streamlines). Layers slide smoothly over one another.

Example: March past of army, honey flowing from a bottle
Characteristics:

- Smooth, orderly motion
- Low Reynolds number (Re ¡ 2000)
- Viscous forces dominate



Laminar Flow

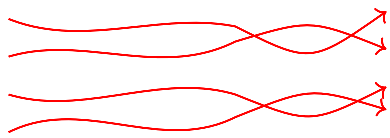
7.3.2 Turbulent Flow

Definition

Fluid particles flow in random, chaotic fashion with no specific path. Irregular fluctuations and mixing.

Example: Passengers at railway platform, rapids in a river
Characteristics:

- Chaotic, irregular motion
- High Reynolds number (Re ¿ 4000)
- Inertial forces dominate



Turbulent Flow

7.4 4. Compressible and Incompressible Flow

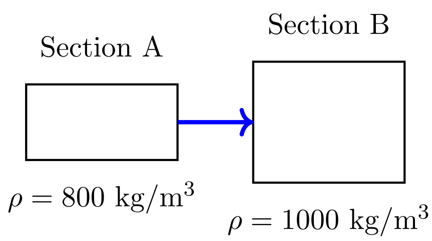
7.4.1 Compressible Flow

Definition

Flow in which fluid density varies from point to point.

$$\rho \neq \text{constant}$$

Example: Flow of air through compressor, supersonic flow



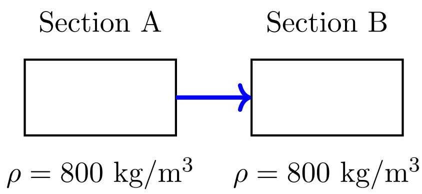
7.4.2 Incompressible Flow

Definition

Flow in which fluid density remains constant throughout.

$$\rho = \text{constant}$$

Example: Most liquid flows (water, oil)



8 Continuity Equation

8.1 Discharge (Q)

Definition

Volume of fluid flowing through a section per unit time. Also called flow rate.

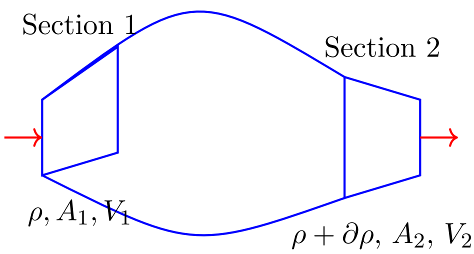
$$Q = A \cdot V$$

where:

- Q = Discharge (m^3/s or L/s)
- A = Cross-sectional area (m^2)
- V = Average velocity (m/s)

8.2 Derivation of Continuity Equation

Consider a control volume with fluid entering at section 1 and leaving at section 2:



Mass flow rate entering section 1: $\dot{m}_1 = \rho A_1 V_1$
Mass flow rate leaving section 2: $\dot{m}_2 = (\rho + \partial\rho)(A + \partial A)(V + \partial V)$
Mass accumulation between sections: $\frac{dm}{dt} = \rho AV - (\rho + \partial\rho)(A + \partial A)(V + \partial V)$
Expanding and neglecting higher-order terms: $\frac{dm}{dt} = -[\rho A\partial V + AV\partial\rho + \rho V\partial A]$
For steady flow: $\dfrac{dm}{dt} = 0$
$\rho A\partial V + AV\partial\rho + \rho V\partial A = 0$
In differential form: $\partial(\rho AV) = 0$
Integrating:

$$\boxed{\rho AV = \text{constant}}$$

8.3 Continuity Equation for Steady Flow

General form (compressible): $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$
For incompressible flow ($\rho_1 = \rho_2$)**:** $\boxed{A_1 V_1 = A_2 V_2 = Q}$
This states: *Discharge remains constant for steady, incompressible flow.*

8.4 Physical Meaning

The continuity equation is based on the **principle of conservation of mass**:

- Mass cannot be created or destroyed
- What flows in must flow out (for steady flow)
- If area decreases, velocity must increase (and vice versa)

8.5 Solved Example

Example

A pipe has diameter 200 mm at section 1 and 300 mm at section 2. Water flows at 4 m/s at section 1. Find:

1. Discharge through the pipe
2. Velocity at section 2

Given:

- $D_1 = 200$ mm $= 0.2$ m
- $D_2 = 300$ mm $= 0.3$ m
- $V_1 = 4$ m/s

Solution:
Step 1: Calculate areas $A_1 = \frac{\pi D_1^2}{4} = \frac{\pi(0.2)^2}{4} = 0.0314\ \text{m}^2$
$A_2 = \frac{\pi D_2^2}{4} = \frac{\pi(0.3)^2}{4} = 0.0707\ \text{m}^2$
Step 2: Calculate discharge $Q = A_1 V_1 = 0.0314 \times 4 = 0.1256\ \text{m}^3/\text{s}$
Step 3: Calculate velocity at section 2
Using continuity equation: $A_1 V_1 = A_2 V_2$
$V_2 = \frac{A_1 V_1}{A_2} = \frac{0.0314 \times 4}{0.0707} = 1.78$ m/s
Answers:

- Discharge: $Q = 0.1256\ \text{m}^3/\text{s} = 125.6$ L/s
- Velocity at section 2: $V_2 = 1.78$ m/s

Note: As the pipe diameter increases, velocity decreases to maintain constant discharge.

9 Quick Reference Summary

Essential Concepts

Newton's Law of Viscosity: $\tau = \mu \dfrac{du}{dy}$

Flow Classification:

- **Steady:** $\dfrac{\partial}{\partial t} = 0$ vs **Unsteady:** $\dfrac{\partial}{\partial t} \neq 0$
- **Uniform:** $\dfrac{\partial}{\partial s} = 0$ vs **Non-uniform:** $\dfrac{\partial}{\partial s} \neq 0$
- **Laminar:** Re < 2000 vs **Turbulent:** Re > 4000
- **Incompressible:** $\rho = \text{const}$ vs **Compressible:** $\rho \neq \text{const}$

Continuity Equation: $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$
For incompressible flow: $A_1 V_1 = A_2 V_2 = Q$
Discharge: $Q = A \cdot V$

10 Comparison: Lagrangian vs Eulerian

Aspect	Lagrangian Method	Eulerian Method
Focus	Individual particles	Fixed points in space
Tracks	Path of particles	Field variables at locations
Coordinates	Material/particle coordinates	Spatial coordinates
Mathematical form	$\vec{r} = F(\vec{r}_0, t)$	$\vec{V} = f(x, y, z, t)$
Applications	Particle dynamics, trajectories	Most fluid mechanics problems
Practicality	Difficult for many particles	More practical for engineering

11 Properties Summary Table

Property	Unit	Definition
Density (ρ)	kg/m^3	Mass per unit volume
Specific Gravity	-	Ratio of density to water density
Viscosity (μ)	N·s/m^2	Resistance to shear deformation
Surface Tension	N/m	Force at free surface due to cohesion
Compressibility	m^2/N	Volume change per unit pressure

Formula Sheet for Quick Revision

Viscosity: $\tau = \mu \frac{du}{dy}$
Discharge: $Q = A \times V$
Area of circular pipe: $A = \frac{\pi D^2}{4}$
Continuity (incompressible): $A_1 V_1 = A_2 V_2$
Speed from velocity components: $V = \sqrt{u^2 + v^2 + w^2}$
Velocity field (Eulerian): $\vec{V} = u\hat{i} + v\hat{j} + w\hat{k} = f(x, y, z, t)$

— *End of Unit III Notes* —